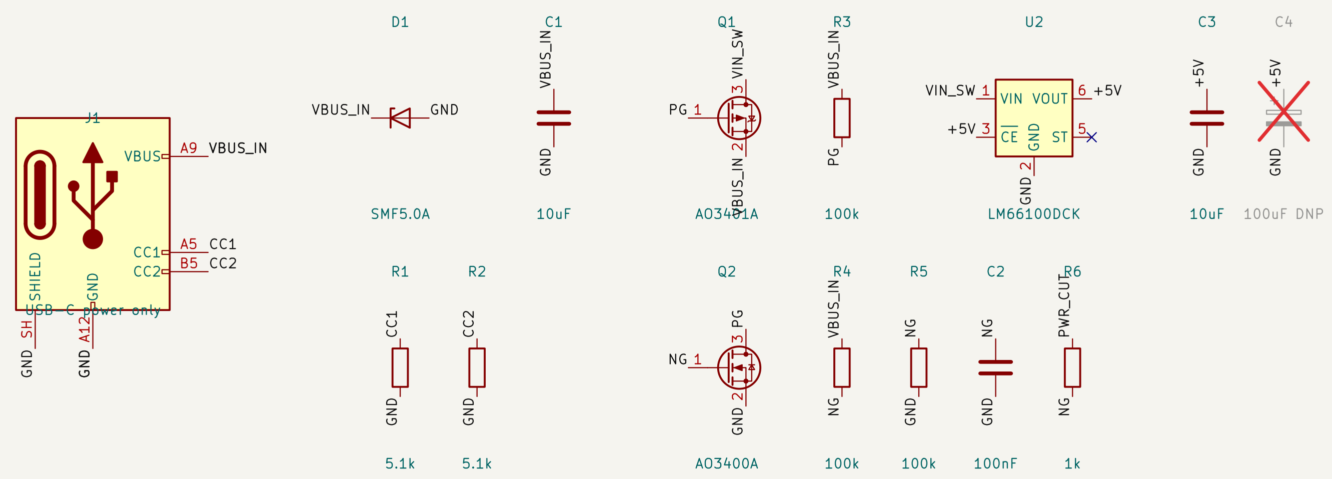
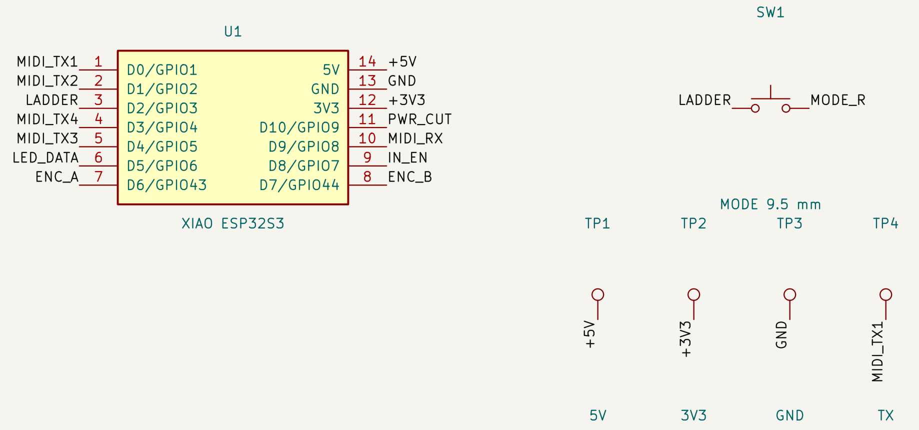
POWER IN: J1 power only (D+/D− not fitted). Q1 cut (on by default), U2 ideal diode onto the XIAO's 5V

U2 ~CE tied to VOUT: the datasheet's ideal–diode mode (blocks current back into J1). The XIAO's 5V pin IS its USB–C VBUS (no part between, Seeed v1.2 schematic): a computer on the XIAO port and a charger on J1 share +5V. Firmware pulls PWR_CUT low once a computer has enumerated it (device mode): Q2 off, Q1 off, J1 cut. At reset PWR_CUT floats: R4/R5 hold Q2 on, so a charger alone always starts it.
C4 (100 uF) not fitted: host–mode hot–plug reserve, against USB's 10 uF limit for a device.

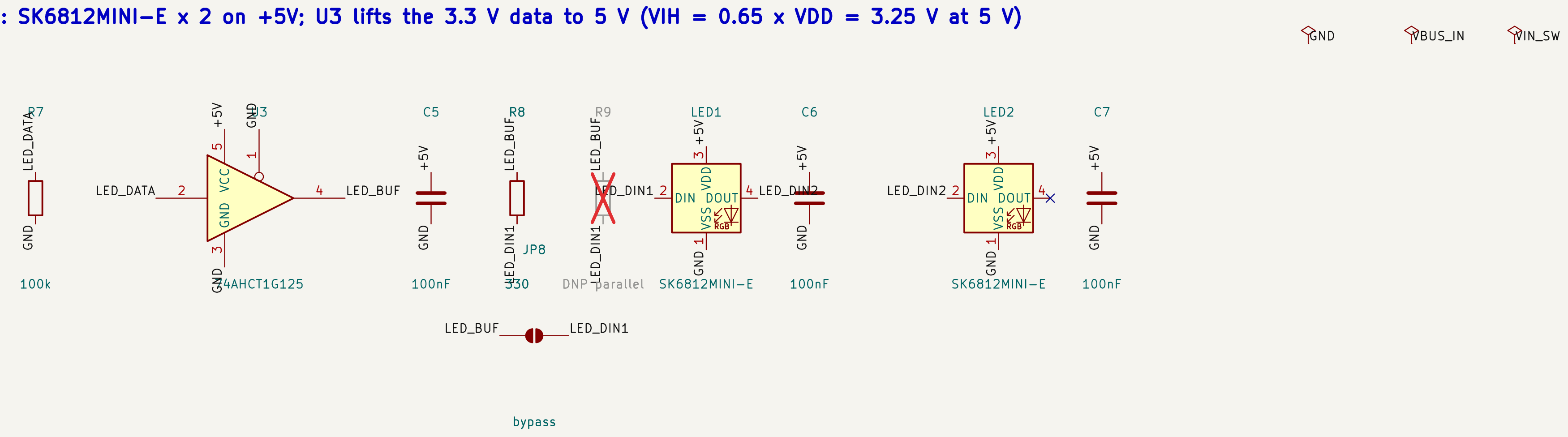


XIAO ESP32S3 (Seeed footprint, 2 x 7 headers or castellations)

D6 = GPIO43 has 499 ohm on the module (U0TXD). D2 = GPIO3 strapping, ignored by default eFuses. ENC_A = D6 = GPIO43, ENC_B = D7 = GPIO44 (U0TXD / U0RXD), IN_EN = D8 = GPIO7, MIDI_RX = D9 = GPIO8.
MIDI_TX1..4 = D0 D1 D4 D3 = GPIO1 2 5 4: UART TX routed to all four by the GPIO matrix (firmware).
LADDER = D2 = GPIO3 (ADC1): MODE / PLAY / PUSH on one resistor ladder. LED_DATA = D5 = GPIO6. PWR_CUT = D10 = GPIO9, low = J1 cut.



LEDS: SK6812MINI–E x 2 on +5V; U3 lifts the 3.3 V data to 5 V (VIH = 0.65 x VDD = 3.25 V at 5 V)



On the board each pair is ONE footprint (J6..J9; long board jank:MIDI_OUT_DIN–504B–M10_or_PJ–320A_AB: the jack's pads T, R, S carry J2..J5's nets, its switch pin SW open; square board: numbered 5, 4, 2 inside the DIN's). J2..J5 are BOM lines only.

MIDI OUT 1–4 (CA–033 at 3.3 V): per output fit ONE of the two, the TRS (type A) OR the DIN–5

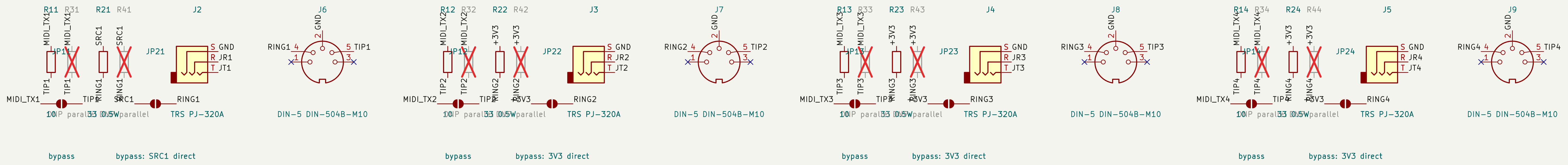
TRS type A: tip = data (10 ohm to the TX pad), ring = source (33 ohm 0.5 W to 3V3), sleeve = GND. DIN–5: pin 5 = tip net, pin 4 = ring net, pin 2 = GND, pins 1 and 3 open.

OUT1: J2 TRS or J6 DIN

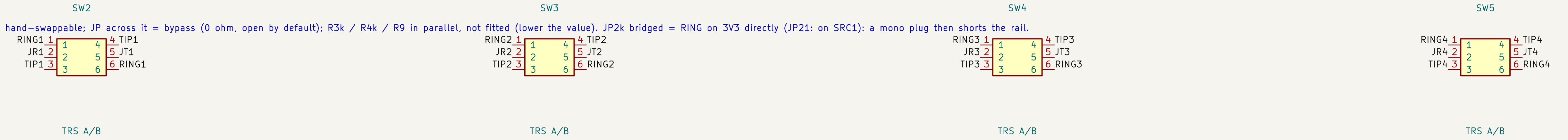
OUT2: J3 TRS or J7 DIN

OUT3: J4 TRS or J8 DIN

OUT4: J5 TRS or J9 DIN



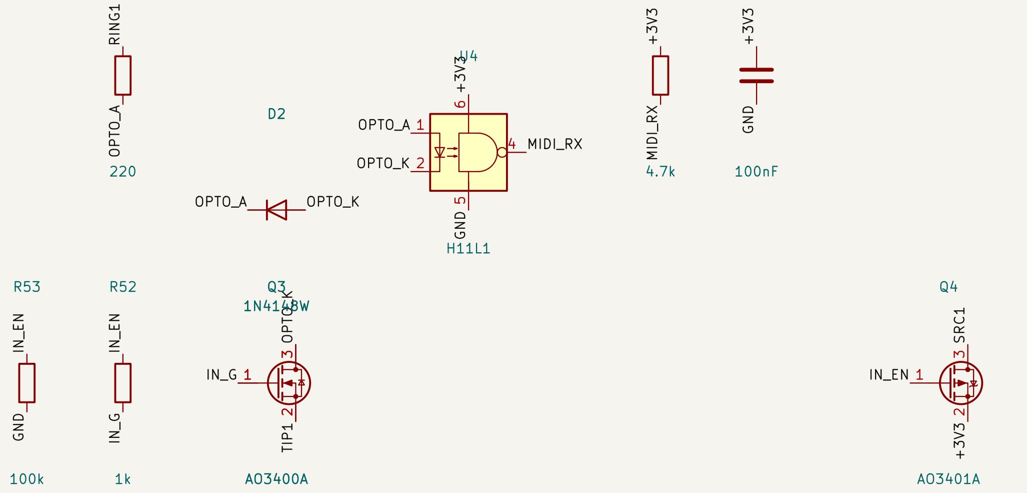
Each MIDI resistor (and R8): axial, hand–swappable; JP across it = bypass (0 ohm, open by default); R3k / R4k / R9 in parallel, not fitted (lower the value). JP2k bridged = RING on 3V3 directly (JP21: on SRC1): a mono plug then shorts the rail.



SW2..SW5: TRS type A / B per output (face side): they swap the jack's T and R between TIPk (data, sink) and RINGk (source) after the resistors; the DIN keeps pin 5 = TIPk, pin 4 = RINGk. 1–2 / 4–5 closed = type A.

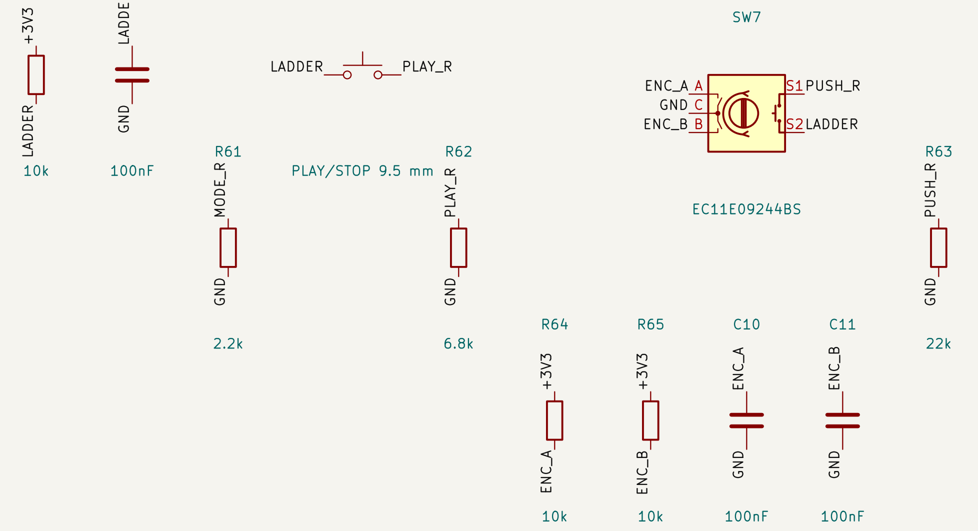
OUT1 AS MIDI IN (software): TX1 to input (Hi–Z), IN_EN high (Q4 off: no source), Q3 on (MIDI_RX = UART RX)

OUT mode: IN_EN low (R53 at reset): Q4 on (3V3 to the 33 ohm), Q3 off, the opto's loop open (its body diode blocks). [illegible]



CONTROLS (face side): ADC ladder on GPIO3: R60 10k to 3V3; MODE 2.2k, PLAY 6.8k, PUSH 22k to GND

Idle 3.30 V, MODE 0.60 V, PLAY 1.34 V, PUSH 2.27 V (one switch at a time). Encoder A/B: 10k to 3V3, 100 nF to GND each.



generated by make_schematic.py; do not edit by hand
hardware/pcb/README.md: the power path, the pin table and every unverified item

Sheet: /
File: jank–pcb.kicad_sch

Title: jank: XIAO ESP32S3 MIDI clock stick

Size: A2 | Date: 2026–11–08 | **Rev: 0.1 schematic for review**
KiCad E.D.A. 10.0.8 | Id: 1/1